



BROMODICHLOROMETHANE

CAS No. 75-27-4

CHBrCl₂ Molecular weight 163.83

Synonym: Dichlorobromomethane

ABSTRACT

Bromodichloromethane (99% pure), one of several trihalomethanes commonly formed after chlorination of water, was selected for study because no carcinogenicity data were available for this compound and because chloroform, a related trihalomethane, had been found to cause tumors in rodents. The general population might be exposed to bromodichloromethane in drinking water supplies, in swimming pools, and in a variety of food substances.

Single-administration, 14-day, 13-week, and 2-year studies were conducted in F344/N rats and B6C3F₁ mice. The chemical was administered by gavage in corn oil because human exposure is primarily oral. Additional studies were performed to evaluate the potential for genetic damage in bacteria and mammalian cells.

Results of the Short-Term Studies: In the single-administration studies, the chemical was administered at doses of 150-2,500 mg/kg per day. All rats and female mice at 1,250 and 2,500 mg/kg and all male mice at 600, 1,250, and 2,500 mg/kg died; 2/5 male rats, 1/5 female rats, and 2/5 female mice at 600 mg/kg died; all animals at lower dose levels survived.

In the 14-day studies, rats received doses of 38-600 mg/kg, and mice received doses of 19-300 mg/kg per day. One female rat at 38 mg/kg and one female rat at 600 mg/kg died. Weight loss or decreased weight gain was seen at 300 and 600 mg/kg in male and female rats. All male mice at 150 and 300 mg/kg died, and one female mouse at 300 mg/kg died; no weight effects were observed in surviving mice. Dose-related necropsy findings included reddened renal medullae in male rats at 600 mg/kg and in male mice at 150 and 300 mg/kg. Clinical signs seen in high dose groups after dosing were hyperactivity in rats and lethargy in mice.

In the 13-week studies, male and female rats received doses of 19-300 mg/kg per day, male mice received doses of 6.25-100 mg/kg per day, and female mice received doses of 25-400 mg/kg per day. Five of 10 male rats and 2/10 female rats at 300 mg/kg died. None of the mice died. Final body weights of male and female rats at 150 and 300 mg/kg were lower than those of vehicle controls (45%-88% of vehicle control weights); final body weights of male mice at 100 mg/kg and female mice at 400 mg/kg were 92% and 94% of those of the vehicle controls. Centrilobular degeneration in the liver and degeneration and necrosis of the kidney were seen in male rats at 300 mg/kg; centrilobular degeneration was seen in female rats at 300 mg/kg; degeneration and necrosis of the kidney were seen in male mice at 100 mg/kg, and centrilobular degeneration of the liver was seen in female mice at 200 and 400 mg/kg.